# magneto

## _travelers guide _for mastering magnetism

| 4 | 3 | 2 |
|---|---|---|
| playback heads | modes | min of tape |

[ tap ] press  [ tap ] hold  › then  noon = 12 o'clock  cw / ccw = clockwise / counter

four playback heads, one record head · echo, looper, phrase sampler, spring reverb, oscillator

magneto · 28 hp · manual rev b

lavender = hold

## 01 buttons · LEDs

mg_01

| | |
|---|---|
| [ modes ] | echo ● › loop ● › sample ● › echo … |
| [ tap ] ×2 | delay time at head 4, 50 ms–15 s · loop / sample: splice in › out › clear (04) |
| [ 1 ]–[ 4 ] | ● green = that head feeds back into the record head, even with its level at zero |
| [ transport ] | the four buttons turn ● red and become ∞ ◂▸ ▸‖ ‖ (06) |

## 02 knobs

mg_02

| | |
|---|---|
| [ speed / pitch ] | tape speed, 8:1 · min ¼ · about 9 o'clock ½ · noon normal · max 2× (fig.04) |
| [ dry ] | your input to the outputs |
| [ wet ] | all four heads to the outputs, after their levels |
| [ rec lvl ] | into the record head · clean to fat and saturated |
| [ repeats ] | feedback for every head whose button is green |
| [ level 1–4 ] | each head's volume · [ ● n ] + turn its knob = pan it (pan on center) |

## 03 tape mechanics

mg_03

| | |
|---|---|
| [ low cut ] | min = full lows · max = thin, peaky "magnetic drum" repeats |
| [ tape age ] | min = fresh tape · cw = warmer · max = a peak that blooms at high repeats |
| [ crinkle ] | friction, creases, splices, dirt · min clean · max chewed |
| [ wow & flutter ] | ccw = a tuned machine · cw = one that needs service |
| [ spring ] | spring reverb level · dry and wet both feed it |

crinkle and wow & flutter bite much harder at slow speeds

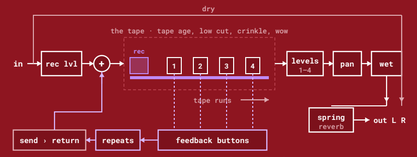
## 05 switches

mg_05

| | |
|---|---|
| [ heads ] | **even** 1/16 · 1/8 · dotted 1/8 · 1/4 · **triplet** 1/16t · 1/8t · 1/4t · 1/4 · **shift** pitched heads (fig.02) |
| [ pan ] | **LRLR** 1 3 left · 2 4 right · **center** all middle, custom pans · **LRRL** 1 4 left |
| one output | everything sums to mono at that jack |

## __fig.01 : signal path

mono in, stereo out




only heads with a green button feed back, even with their level at zero
loop / sample: head 4 sits before the record head and plays the loop

## 04 modes

mg_04 · [ modes ]

| | |
|---|---|
| ● echo | **four-head tape delay** · [ tap ] sets the head spacing, [ speed ] the tape speed |
| ● loop | **sound-on-sound** · head 4 plays the loop, heads 1–3 keep echoing · needs [ 4 ] green and level 4 up |
| loop [ tap ] | splice in › splice out (plays) › clear · [ repeats ] lower = the loop fades as you layer |
| ● sample | **phrase sampler** · [ tap ] record › stop › clear · [ ▸| ] plays it once |
| length | longest loop or sample: 15 s at max speed · 30 s noon · 1 min half · 2 min min |
| heads 1–3 | keep the delay time you had when you entered loop / sample, or follow [ clk in ] |

## __fig.04 : the speed dial

8:1




**slower**
longer, lower, darker · wow and crinkle grow

**faster**
shorter, higher, cleaner

**tap instead**
moves the heads: new delay, same pitch

**turning it**
repitches what's on the tape

## 09 audio

mg_09 · 20 vpp

| | |
|---|---|
| mono in | [ left in ] only |
| mono out | [ left out ] only · all sums to it |
| stereo out | both outs · works from a mono in too |
| [ send ] [ return ] | fx loop before the record head · both patched = repeats go mono |
| the module | 28 hp · 41 mm · 210 mA each rail · 96 khz |

## __fig.02 : head spacing

and the clk outs




tapped or clocked at 1000 ms · clk 1–4 out tick at these times, 50% duty, phase-aligned
shift = each head plays at its own speed · loop / sample: clk 4 out = the loop length

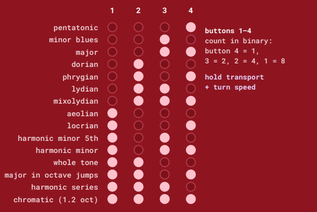
## 07 pitch quantize

mg_07 · [ tap ]

| | |
|---|---|
| [ tap ] | [ speed / pitch ] snaps to the notes of a scale · tap LED ● |
| [ transport ] + [ speed ] | turn to pick one of 15 scales · buttons 1–4 show which (fig.05) |
| range | two octaves (4:1) · chromatic 1.2 octaves · speed cv snaps to half steps |

## __fig.05 : the 15 scales

[ transport ] + [ speed ]




## 08 self-oscillation

mg_08 · no input needed

1. [ speed / pitch ] fully ccw · [ repeats ] at 100%
2. [ tap ] ×2, as fast as you can · it starts to sing
3. play [ speed ] and the four tape knobs · it wanders through pitches
4. [ transport ] › [ ∞ ] locks the pitch · speed and speed cv still play it

speed cv is 1 v/oct, −3 to +3 v: six octaves, three below the knob alone · LRLR / LRRL widen the tone

## 12 dip switches

mg_12 · on the back, off by default

| | |
|---|---|
| s1 [ feedback cv ] | on: with the transport LED off, the transport jacks toggle feedback buttons 1–4 instead |
| s2 [ dual split ] | on: left in / out = mono four-head delay · right in / out = mono spring reverb |

## 06 transport

mg_06 · [ transport ] first

| | |
|---|---|
| [ ∞ ] infinite | record head off · the last delay cycle or loop plays forever |
| [ ◂▸ ] reverse | from that moment · echo: the whole tape backwards · loop / sample: the loop backwards |
| [ ▸| ] restart | loop / sample from the top · realigns shift heads · leaves pause instantly |
| [ ‖ ] pause | the tape slows to a stop, or spins back up · while paused, level 4 = how fast |
| scrub | while paused, [ speed / pitch ] scrubs about a foot of tape · slower speed = more audio |
| defaults | all latch except ▸|, always a trigger · change one at power on (mg_15) |

## 10 cv inputs

mg_10

**continuous, −5 to +5 v** — the knob is the offset

| | |
|---|---|
| [ speed ] | −3 to +3 v, 1 v/oct |
| [ wet ] [ repeats ] [ spring ] | add to their knobs |

**triggers, 0–5 v rising edge**

| | |
|---|---|
| [ clk in ] | delay time as a quarter note, 50 ms–15 s |
| [ rec gate ] | record head on / off · rec lvl LED ● when muted |
| [ shift ] | shift head mode on / off |
| [ ∞ ] [ ◂▸ ] [ ▸| ] [ ‖ ] | the transport buttons |
| [ tap ] | the tap button: delay, splice or sample · 10 ms–15 s |

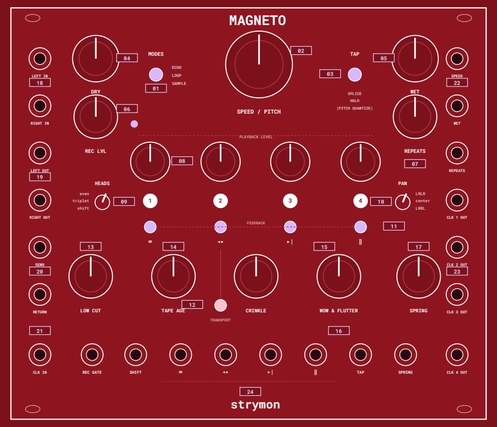
## 11 tricks

mg_11 · from the manual

| | |
|---|---|
| sub-octave | shift heads · a gate into [ ▸| ] cv = a zero-latency sub-octave from head 3 |
| endless | sample mode · [ clk 4 out ] › [ ▸| ] cv retriggers the phrase |
| extra tone | [ ∞ ] on a very short delay adds a harmonic at the seam |

## __fig.03 : front panel

01–17 = controls · 18–24 = jacks




| | |
|---|---|
| [ 01 ] __modes : | echo · loop · sample |
| [ 02 ] __speed / pitch : | tape speed 8:1 |
| [ 03 ] __tap : | delay · splice · hold = quantize |
| [ 04 ] __dry : | input level |
| [ 05 ] __wet : | all heads |
| [ 06 ] __rec lvl : | into the tape · LED |
| [ 07 ] __repeats : | feedback |
| [ 08 ] __level 1–4 : | each head |
| [ 09 ] __heads : | even · triplet · shift |
| [ 10 ] __pan : | LRLR · center · LRRL |
| [ 11 ] __1–4 : | feedback · transport |
| [ 12 ] __transport : | flips 11 to red |

| | |
|---|---|
| [ 13 ] __low cut : | thins the repeats |
| [ 14 ] __tape age : | darker, peakier |
| [ 15 ] __crinkle : | damaged tape |
| [ 16 ] __wow & flutter : | speed wobble |
| [ 17 ] __spring : | reverb level |
| [ 18 ] __ins : | left = mono |
| [ 19 ] __outs : | one = mono sum |
| [ 20 ] __send · return : | before record |
| [ 21 ] __clk in : | quarter note |
| [ 22 ] __cv : | speed · wet · repeats |
| [ 23 ] __clk 1–4 out : | the heads |
| [ 24 ] __gates : | rec · shift · transport · tap |

## _where things hide

no menus · mg_15

### held buttons

- tap pitch quantize
- transport + speed pick a scale
- green n + level n pan head n (pan center)
- paused: level 4 slowdown speed

### at power on

- hold ∞ ◂▸ ▸| or ‖ flips it: momentary ● / latch ●
- dip switches s1 feedback cv · s2 dual split (12)

### LED codes

| | |
|---|---|
| modes · tap | ● echo ● loop ● sample ● tap = quantize |
| buttons 1–4 | ● feedback on ● transport layer / scale code |
| rec lvl | ● none ● clean ● saturating ● heavy |